

CONCEPT MAP 2: EVOLUTION OF CLOUD COMPUTING

Project Title: Architectural Analysis and Deployment of Serverless Computing in Modern Cloud Platforms

EVOLUTION OF CLOUD COMPUTING

1. HARDWARE EVOLUTION

- Mainframe Computing
- Client-Server Computing
- Virtualized Servers
- Cloud Data Centers
- Modern Cloud Platforms

2. INTERNET & SOFTWARE EVOLUTION

- Static Websites
- Dynamic Web Applications
- Web Services (SOAP)
- REST APIs
- Cloud-Native Applications
- Serverless Applications

3. DISTRIBUTED SYSTEMS EVOLUTION

- Cluster Computing
- Grid Computing
- Utility Computing
- Cloud Computing
- Microservices
- Event-Driven Serverless

4. CLOUD SERVICE EVOLUTION

- Traditional Infrastructure
- IaaS
- PaaS
- SaaS
- FaaS (Serverless)
- XaaS

5. EVOLUTION DRIVERS

- Virtualization
- High-Speed Internet
- Automation
- Containers
- APIs
- Elastic Scaling

6. KEY OUTCOMES

- Scalability
- Elasticity
- Cost Efficiency
- High Availability
- Rapid Deployment
- Global Accessibility

7. CONNECTION TO THE PROJECT

- The evolution of cloud computing leads to serverless architectures that enable building scalable, cost-effective, and highly available applications with minimal operational overhead.

SERVERLESS COMPUTING (FaaS)

ARCHITECTURAL ANALYSIS AND DEPLOYMENT OF SERVERLESS COMPUTING IN MODERN CLOUD PLATFORMS